

HCSM-01

The 8×8 Torus Substrate from $d = 4$

The Framework Relation $N = d \cdot 2^d$, the Emergent Dimension, and the Selection of $L = 8$

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Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. This paper establishes the substrate itself.

The *substrate-level disclosed inputs* are three: (i) the substrate is a periodic square lattice with D_4 point-group symmetry; (ii) the operators of interest are nearest-neighbour and positive-semidefinite with kernel equal to the constant functions; (iii) the emergent spacetime dimension equals $\dim \ker D$ of the substrate's Hodge–Dirac operator. Given these, the paper derives:

1. the 5-point Laplacian is the unique such operator on Λ (§2.2);
2. the emergent spacetime dimension is $d = 4$, via the Hodge–Dirac kernel on the 2-torus (§3.3, citing HCSM-03 for the signature);
3. the framework relation $N = d \cdot 2^d$ is a consistency identity at $d = 4$, giving $N = 64 = 8^2$, hence $L = 8$ (§4.1–§4.2);
4. the substrate size $L = 8$ is independently selected by two structurally distinct derivations: equal entropy spacing on the torus Laplacian (§4.3) and maximal cancellation of the winding generator (§4.4, HCSM-14.b);
5. the multiplicity at the self-paired eigenvalue $\lambda = 4$ is $2(L - 1) = 14$ (§4.5).

The paper closes with an explicit *disclosed-input ledger* (§5) that separates substrate-level disclosures from physics-level disclosures (identification thesis, charge table, mass labels) that live in downstream papers. All numerical claims are verified at machine precision.

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1 Introduction

1.1 What HCSM-01 establishes

HCSM derives the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Every downstream result depends on the substrate being what it is. This paper establishes the substrate and states exactly what is derived and what is disclosed.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

1.2 Disclosed input set at the substrate level

Three items are disclosed at the substrate level. They are the primitive assumptions of the framework, and every substrate-level theorem in the HCSM stack depends on them.

(S1) Square-lattice substrate. The substrate is the periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with the D_4 point group of the square acting as its symmetry group.

(S2) Locality and positivity. The scalar Laplacian of interest is nearest-neighbour, positive-semidefinite, and annihilates exactly the constant functions.

(S3) Emergent dimension. The emergent spacetime dimension equals $\dim \ker D$, where $D = d + d^*$ is the Hodge–Dirac operator of the substrate’s Hodge complex.

These three disclosures are the substrate-level anchor of the framework. Nothing else is disclosed at the substrate level. §2.2 shows that (S1) and (S2) force the 5-point Laplacian; §3 shows that (S3), together with the Hodge star on $\ker D$ (HCSM-03), forces $d = 4$ and the Lorentzian signature.

1.3 What is derived here, what is derived elsewhere

Derived in this paper:

- The 5-point Laplacian, from (S1) and (S2) (Theorem 2.1).
- $\dim \ker D = 4$ via the Betti numbers of T^2 (Theorem 3.1).

- The framework relation $N = d \cdot 2^d$ as a consistency identity at $d = 4$ (Consistency 4.1).
- The multiplicity at $\lambda = 4$: $2(L - 1) = 14$ at $L = 8$ (Theorem 4.4).

Derived in companion papers:

- $d = 4$ from the Hodge star on $\ker D$: HCSM-03.
- $L = 8$ from equal entropy spacing: companion equal-entropy paper.
- $L = 8$ from maximal cancellation of the winding generator: HCSM-14.b.
- The Hodge complex and $D^2 = \Delta$: HCSM-02.

1.4 Cross references

Depends on: HCSM-00, HCSM-02, HCSM-03, HCSM-04, HCSM-14.b.

Propagates to: every paper in the HCSM stack that uses Λ , $N = 64$, $L = 8$, $d = 4$, or the 14-mode multiplet.

2 The substrate

2.1 The periodic $L \times L$ torus

The substrate is the periodic square lattice

$$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2,$$

of N sites arranged in an $L \times L$ grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; topologically, Λ_L is the torus $T^2 = S^1 \times S^1$. The D_4 point group of the square acts by the rotations and reflections of the lattice.

2.2 The 5-point Laplacian is forced

Theorem 2.1 (Laplacian uniqueness). *Let $\Delta : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator on the substrate $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ satisfying:*

1. **D_4 -invariance:** $[R, \Delta] = 0$ for every $R \in D_4$.
2. **Nearest-neighbour locality:** $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbours of x .
3. **Positivity and kernel:** Δ is positive semidefinite and $\ker \Delta$ is exactly the set of constant functions.

Then Δ is uniquely determined up to overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1). \quad (1)$$

Proof. Condition (2) writes $\Delta f(x) = a_0 f(x) + \sum_{\mu \in \{\pm e_1, \pm e_2\}} a_\mu f(x + \mu)$. Condition (1) forces $a_\mu = a$ for all four nearest-neighbour directions (the four $\pm e_1, \pm e_2$ directions are in a single D_4 orbit). So $\Delta f(x) = a_0 f(x) + a \sum_\mu f(x + \mu)$. Condition (3) forces $\Delta \mathbf{1} = 0$, hence $a_0 = -4a$. Normalize $a = 1$. $\square$

Remark 2.2. The 5-point Laplacian is annihilated by the constant functions by construction; its eigenvalue on the plane wave $\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/L}$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad n_1, n_2 \in \{0, \dots, L - 1\}. \quad (2)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$.

2.3 The Hodge complex

The Hodge complex of differential forms on the 2-torus Λ_L has three graded pieces:

$$\Omega^0 = \mathbb{R}^N, \quad \Omega^1 = \mathbb{R}^{2N}, \quad \Omega^2 = \mathbb{R}^N, \quad (3)$$

with total dimension $N + 2N + N = 4N$. On $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ this gives total dimension $4 \cdot 64 = 256$.

By HCSM-02, the Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree, where Δ is the 5-point Laplacian (1).

3 The emergent dimension and its signature

3.1 The kernel of the Hodge–Dirac operator

Theorem 3.1 (Kernel dimension). *On the 2-torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the Hodge–Dirac operator $D = d + d^*$ has kernel isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}),$$

with dimension $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.

Proof. By the discrete Hodge theorem (Dodziuk 1976), $\ker D$ on a discrete torus is isomorphic to the de Rham cohomology. The Betti numbers of T^2 are $(b_0, b_1, b_2) = (1, 2, 1)$. $\square$

The four harmonic forms of $\ker D$ are explicit: the constant 0-form v_0 (spanning H^0); the two independent harmonic 1-forms v_1^A, v_1^B (spanning H^1); the constant 2-form v_2 (spanning H^2).

3.2 The Lorentzian signature

By disclosure (S3), the emergent spacetime dimension equals $\dim \ker D = 4$. The signature of the emergent metric is derived in HCSM-03 from the Hodge star restricted to $\ker D$. We summarize that result.

Theorem 3.2 (Lorentzian signature, HCSM-03). *The Hodge star $\star$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the ± 1 eigenvalues come from the $(H^0 \oplus H^2)$ sector, the $\pm i$ eigenvalues from the H^1 sector. The intersection form $Q(\alpha, \beta) = \int \alpha \wedge \beta$ on the even-degree sector and the Hodge inner product $\langle \alpha, \beta \rangle = \int \alpha \wedge \star \beta$ on the odd-degree sector combine into a canonical metric of signature $(3, 1)$ on $\ker D$. The timelike direction is $(v_0 + v_2)/\sqrt{2}$.*

Together, Theorems 3.1 and 3.2 give $d = 4$ with signature $(-, +, +, +)$, both derived from the substrate and the Hodge star, with no external input and no Wick rotation.

Remark 3.3 (Two independent routes to $d = 4$). The number $d = 4$ admits two complementary derivations:

1. The internal Hodge-star argument (Theorem 3.2): the Hodge star on $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$; the intersection form and Hodge inner product force signature $(3, 1)$.
2. The dimension-counting argument: $\dim \ker D = 4$ is the fixed point of the torus Betti-number structure under Hodge duality. The identity $b_0 = b_2$ and $b_1 = 2$ are the only values compatible with Hodge duality on a connected closed surface without boundary.

The two routes agree, and both are derived from the substrate.

4 The substrate size

4.1 The framework relation $N = d \cdot 2^d$ as a consistency identity

The HCSM paper stack uses the relation

$$N = d \cdot 2^d \tag{4}$$

with N the number of substrate sites and d the emergent spacetime dimension. We present (4) as a *consistency identity*, not a derivation. The reason is structural: on the 2-torus the Hodge complex has dimension $4N = 256$, while the abstract exterior algebra on $\mathbb{R}^d$ has dimension $2^d = 16$. These are different mathematical objects; a tensor-product argument identifying them is a plausibility statement, not a theorem.

What (4) does is state that at $d = 4$ the numbers 4 and 2^4 multiply to give the substrate size $N = 64$. This is consistent with the substrate being $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with $L = 8$:

Consistency Relation 4.1 (Substrate size at $d = 4$). At $d = 4$, the framework relation (4) gives $N = 4 \cdot 16 = 64$. Since $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ requires $N = L^2$, this forces $L = \sqrt{64} = 8$.

The physical selection of the substrate size is therefore not (4) alone. It is the two independent derivations of $L = 8$ below.

4.2 The perfect-square observation

At $d = 4$, $N = 64 = 8^2$ is a perfect square, so the substrate is a square lattice. We note that $d \cdot 2^d = L^2$ is not unique to $d = 4$: the integer solutions form the infinite family $d \in \{4k^2 : k \geq 1\}$, with $d = 16, 36, 64, \dots$ also giving perfect squares. The physical selection of $d = 4$ is therefore not from the perfect-square condition; it is from the emergent-dimension argument of §3.

4.3 First derivation of $L = 8$: equal entropy spacing

On the $L \times L$ torus, the standing-wave modes fall into discrete Shannon entropy classes. The companion equal-entropy-spacing paper proves the following.

Theorem 4.2 (Equal entropy spacing selects $L = 8$). *Among all integers $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally spaced Shannon entropy classes is $L = 8$. At $L = 8$ the three classes are 6.0 bits, 5.5 bits, 5.0 bits, with uniform spacing 0.5 bits.*

This gives $L = 8$ independent of the Standard Model particle count.

4.4 Second derivation of $L = 8$: maximal winding cancellation

The winding generator of HCSM-14 generalizes to any even L as $H_{\text{wind}}(L) = i \sin(\pi M/L)$ on the self-paired mode set of Λ_L . HCSM-14.b proves the following.

Theorem 4.3 (Maximal cancellation selects $L = 8$, HCSM-14.b). *Let $\mathcal{G}_L$ be the parity block graph of $H_{\text{wind}}(L)$ on the five nodes $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$. The number of block couplings that vanish identically is*

$$\#\text{zeros}(\mathcal{G}_L) = \begin{cases} 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L \in \{4, 8\}. \end{cases}$$

Among the two maximally-cancelling sizes, only $L = 8$ carries the 14-mode multiplet. At $L = 8$ the surviving five block couplings form a unicyclic graph on $d+1 = 5$ nodes with rank-1 edges; the winding kernel has dimension $d(d+1)/2 = 10$; and the kernel-diagonal values are $\{2/d, (d-1)/d\}$.

The two derivations (§4.3 and §4.4) are structurally independent: the first uses only the torus Laplacian’s entropy classes, the second uses only the winding generator’s parity block structure. Their agreement selects $L = 8$.

4.5 The 14-mode count

Theorem 4.4 (Midpoint multiplicity). *On the $L \times L$ torus with L even, the multiplicity of the self-paired eigenvalue $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(L) = 2(L - 1).$$

At $L = 8$ this gives 14.

Proof. The self-paired eigenvalue is the fixed point of $\lambda \leftrightarrow 8 - \lambda$. Its eigenspace is the set of modes (n_1, n_2) with $\cos(2\pi n_1/L) + \cos(2\pi n_2/L) = 0$, i.e., $n_1 \pm n_2 = L/2 \pmod{L}$. Counting the solutions gives $2(L - 1)$. $\square$

The 14 modes at $\lambda = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ are the subject of HCSM-04 and are identified with one Standard Model generation in HCSM-23.

5 Disclosed-input ledger

We state the full ledger of disclosed inputs. Substrate-level disclosures (§1.2) are the framework’s primitives. Physics-level disclosures live in downstream papers and are listed here for completeness.

5.1 Substrate-level disclosed inputs

- (S1) The substrate is a periodic square lattice with D_4 point-group symmetry.
- (S2) The scalar Laplacian of interest is nearest-neighbour, positive-semidefinite, and annihilates exactly the constant functions.
- (S3) The emergent spacetime dimension equals $\dim \ker D$.

5.2 Physics-level disclosed inputs (forward-referenced)

- (P1) **Identification thesis.** The 14 modes at $\lambda = 4$ are in one-to-one correspondence with the 14 Standard Model particles of one generation. Disclosed in HCSM-23 §3.
- (P2) **Physical identifications of D_4 irreps.** $A_1 \leftrightarrow \text{color}$; $B_1 \leftrightarrow \text{weak isospin}$. Disclosed in HCSM-06 §7.
- (P3) **Physical identification of the orientation flip.** $F \leftrightarrow \text{CP}$. Disclosed in HCSM-43.
- (P4) **Physical identification of spin structure.** $\chi_{\text{spin}} \leftrightarrow L/R$ chirality. Disclosed in HCSM-42.
- (P5) **Physical identification of the winding generator.** $H_{\text{wind}} \leftrightarrow F_{\mu\nu} \tilde{F}^{\mu\nu}$. Disclosed in HCSM-43.
- (P6) **Charge table.** The charge functional $q_3 = F(n_1, \sigma)$, with $\sigma = \text{sign}(s)$ the sign of the zero-state amplitude, is a disclosed input (Input 3 of HCSM-24). The closed form of F is an open item in HCSM-24.
- (P7) **Mass-ordering labels.** The $N/4$ values assigned to the 14 modes are the framework’s empirical anchor for the mass sector. Disclosed in HCSM-23 (Input 3).
- (P8) **Shell values.** The shell values $\{0, 1, 1.5, 2, 3\}$ are inputs from the mass spectrum. Disclosed in HCSM-32.

- (P9) **Base formula for the 5+9 split.** The framework base formula in HCSM-11 is a disclosed input.
- (P10) **Three hidden-sector choices.** The gauge-algebra identification, $U(1)_Y$ neutrality, and fermionic statistics of the B_1 doublets are disclosed inputs in HCSM-38.
- (P11) **Planck mass M_P .** The single dimensionful anchor of the framework.

5.3 Open derivation targets

- The closed form of the charge functional $F(n_1, \sigma)$ (HCSM-24).
- The microscopic derivation of the three hidden-sector choices (HCSM-38).
- The derivation of the identification thesis (P1) from substrate invariants alone.
- The derivation of H_{wind} 's sine form from more primitive principles.
- The discrete-to-continuum scaling limit (item 6.14).

6 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The verification script `HCSM-01_verification.py` constructs the substrate Laplacian on $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ for $L \in \{4, 6, 8, 10, 12, 16\}$, and verifies:

1. The 5-point Laplacian uniqueness (Theorem 2.1).
2. $\dim \ker D = 4$ (Theorem 3.1).
3. $N = 64$ at $d = 4$ (Consistency 4.1).
4. The multiplicity formula $\text{mult}_{\lambda=4}(L) = 2(L - 1)$ (Theorem 4.4).
5. The equal-entropy-spacing selection of $L = 8$.
6. The maximal-cancellation selection of $L = 8$ (HCSM-14.b ledger).
7. The four Betti numbers of T^2 .

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack. The complete stack is at <https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->.

7 Conclusion

This paper has established the substrate of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites.

The substrate-level disclosed inputs are three: a square-lattice substrate with D_4 symmetry (S1); nearest-neighbour, positive-semidefinite locality with kernel equal to the constant functions (S2); and the identification of the emergent spacetime dimension with $\dim \ker D$ (S3).

Given these, the paper derives:

- The 5-point Laplacian is the unique operator satisfying (S1) and (S2) (Theorem 2.1).
- $\dim \ker D = 4$ (Theorem 3.1).
- The Lorentzian signature $(3, 1)$ on $\ker D$, via HCSM-03's Hodge-star argument (Theorem 3.2).
- $d = 4$ (from (S3) and $\dim \ker D = 4$).
- The framework relation $N = d \cdot 2^d$ as a consistency identity at $d = 4$ (Consistency 4.1).

- Two independent derivations of $L = 8$: equal entropy spacing on the torus Laplacian (§4.3) and maximal winding cancellation (§4.4, HCSM-14.b).
- The 14-mode count $2(L - 1) = 14$ at $L = 8$ (Theorem 4.4).

The substrate is not chosen. It is what the framework’s three substrate-level disclosures jointly require: a square lattice with D_4 symmetry, nearest-neighbour positive-semidefinite locality, and emergent dimension dim ker D . Every downstream result of the HCSM stack follows from these plus the physics-level disclosures catalogued in §5.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-03: Lorentzian Signature from the Hodge Star*, HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-06: Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-14.b: Maximal Cancellation Selects $L = 8$* , HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-23: The Identification of the 14 Modes with One Standard Model Generation*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-24: The Charge Theorem*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-38: Hidden Sector Content*, HCSM White Paper Series (2026).
- [11] J. Dodziuk, *Finite-difference approach to the Hodge theory of harmonic forms*, Amer. J. Math. **98**, 79 (1976).